

STA510 - Mandatory assignment 1

Report (theory answers) – autumn 2026

Name: [fill in your name] Student number: [fill in]

This report answers the theory questions of mandatory assignment 1. The accompanying R-code (file mandatory1.R) implements all subproblems marked [R]; console output and figures referenced below are produced by running that script (plots are also collected in output/mandatory1_plots.pdf).

Problem 1

a) Finding the normalising constant c

The density is $f(x; a) = c(x - a)(a + 20 - x)$ for $a \leq x \leq a + 20$, and 0 otherwise. Using the substitution $u = x - a$ (so $x = a$ gives $u = 0$ and $x = a + 20$ gives $u = 20$), the normalisation condition $\int_a^{a+20} f(x; a) dx = 1$ becomes:

$$c \int_0^{20} u(20 - u) du = 1$$

Expanding the integrand and integrating term by term:

$$\int_0^{20} (20u - u^2) du = \left[10u^2 - \frac{u^3}{3} \right]_0^{20} = 10(20)^2 - \frac{20^3}{3} = 4000 - \frac{8000}{3} = \frac{4000}{3}$$

Setting $c \cdot \frac{4000}{3} = 1$ gives the normalising constant:

$$c = \frac{3}{4000}$$

Since $c > 0$ and $f(x; a) \geq 0$ on $[a, a + 20]$ (as $(x - a) \geq 0$ and $(a + 20 - x) \geq 0$ there) and integrates to 1, $f(x; a)$ is a valid pdf.

b) Expectation and variance of X

Write $X = a + U$ where $U = X - a$ has density $g(u) = c u(20 - u)$ on $[0, 20]$ (a scaled Beta(2,2) shape, symmetric about $u = 10$). Using $c = 3/4000$:

$$E[U] = c \int_0^{20} u^2(20 - u) du = c \left[\frac{20u^3}{3} - \frac{u^4}{4} \right]_0^{20} = \frac{3}{4000} \cdot \frac{40000}{3} = 10$$

$$E[U^2] = c \int_0^{20} u^3(20 - u) du = c \left[5u^4 - \frac{u^5}{5} \right]_0^{20} = \frac{3}{4000} \cdot 160000 = 120$$

$$\text{Var}(U) = E[U^2] - (E[U])^2 = 120 - 100 = 20$$

Since $X = a + U$ with a a constant, $E[X] = a + E[U]$ and $\text{Var}(X) = \text{Var}(U)$:

$$E(X) = a + 10 \quad \text{Var}(X) = 20$$

Interpretation: the expected daily maximum wind speed is exactly the midpoint of the feasible range $[a, a + 20]$, which makes sense because $f(x; a)$ is symmetric about $a + 10$ (a parabola that is zero at the endpoints and largest in the middle). The variance, $20 \text{ m}^2/\text{s}^2$ (standard deviation $\sqrt{20} \approx 4.47 \text{ m/s}$), does not depend on a : the threshold a only shifts the location of the distribution, it does not change how spread out the wind speeds are around their mean.

c) Deriving the CDF $F(x; a)$

For $a \leq x \leq a + 20$, using the same substitution $u = t - a$:

$$F(x; a) = \int_a^x f(t; a) dt = c \int_0^{x-a} u(20 - u) du = c \left[10u^2 - \frac{u^3}{3} \right]_0^{x-a}$$

$$F(x; a) = \frac{3}{4000} \left(10(x - a)^2 - \frac{(x - a)^3}{3} \right)$$

which is exactly the form stated in the problem (for $a \leq x \leq a + 20$; $F = 0$ below a and $F = 1$ above $a + 20$).

d) [R] Inverse transform simulation (a=5, n=5000) - see mandatory1.R

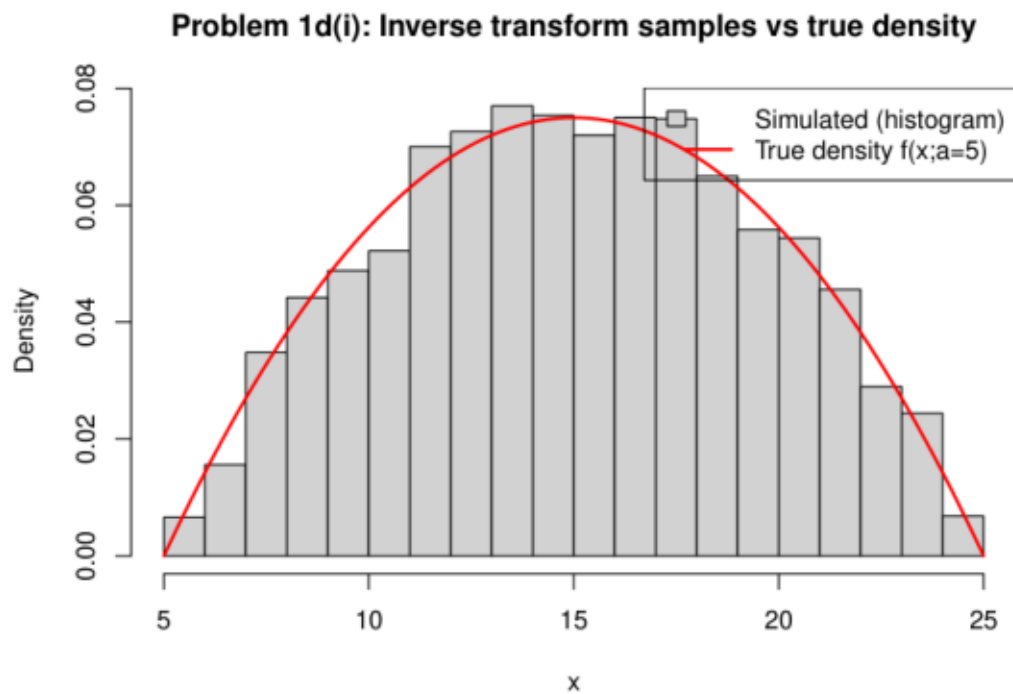


Figure 1: histogram (Scott's rule) of inverse-transform samples vs. true density $f(x; a=5)$.

i) Comment on the fit: the histogram matches the parabolic true density very closely: both are symmetric and unimodal on $[5, 25]$ with a peak around $x = 15$, confirming that the numerical uniroot-based inversion of F correctly reproduces the target distribution.

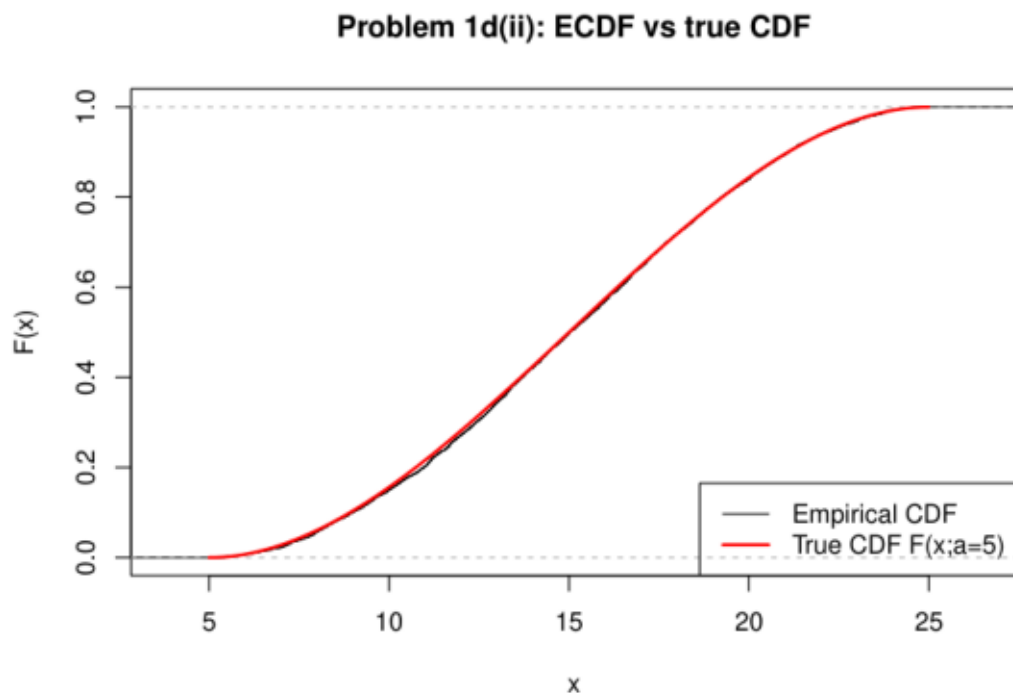


Figure 2: empirical CDF vs. true CDF $F(x; a=5)$.

ii) Comment on the fit: the empirical CDF lies essentially on top of the true CDF over the whole support, with only the small, expected sampling fluctuations of an ECDF based on $n = 5000$ points. This further confirms that the inverse-transform method works correctly.

e) Acceptance-Rejection method with uniform proposal

i) Finding M : with proposal $g(x) = 1/20$ on $[a, a + 20]$, we need the smallest M such that $f(x; a) \leq M g(x)$ for all x , i.e. $M \geq 20 \max_x f(x; a)$ for all x , so $M = 20 \max_x f(x; a)$.

By symmetry $f(x; a)$ is maximised at the midpoint $x = a + 10$ (where $u = x - a = 10$):

$$f(a + 10; a) = c \cdot 10 \cdot 10 = 100c = 100 \cdot \frac{3}{4000} = \frac{3}{40} = 0.075$$

$$M = 20 \cdot 0.075 = 1.5$$

ii) AR algorithm for this f and g :

- 1. Draw a proposal $X \sim \text{Uniform}(a, a + 20)$, i.e. draw $U_1 \sim U(0, 1)$ and set $X = a + 20 U_1$.
- 2. Draw an independent $U_2 \sim U(0, 1)$.
- 3. Compute the acceptance probability $\alpha(X) = f(X; a)/(M g(X)) = f(X; a)/(1.5 \cdot 1/20) = f(X; a)/0.075$.
- 4. If $U_2 \leq \alpha(X)$, accept X as a draw from f ; otherwise reject and return to step 1.
- 5. Repeat until $n = 5000$ values have been accepted.

iii) Theoretical acceptance rate:

$$P(\text{accept}) = \frac{1}{M} = \frac{1}{1.5} = \frac{2}{3} \approx 0.667$$

Expected number of proposals needed to obtain $n = 5000$ accepted samples:

$$n \cdot M = 5000 \cdot 1.5 = 7500$$

f) [R] Acceptance-Rejection simulation (a=5, n=5000) - see mandatory1.R

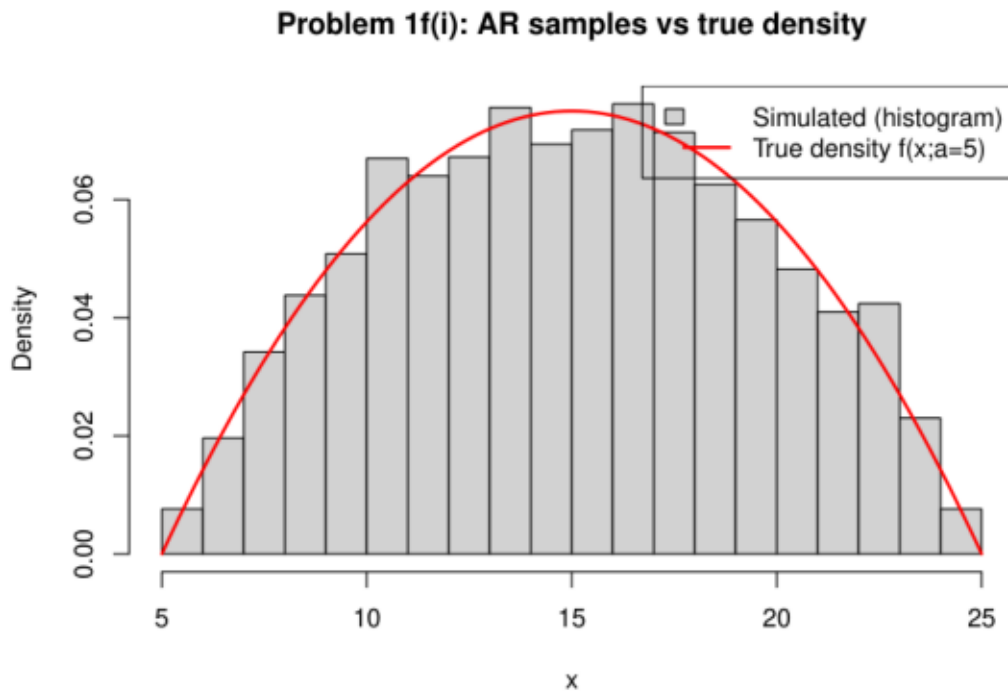


Figure 3: histogram of AR samples vs. true density $f(x; a=5)$.

- i) Comment on the fit: as with the inverse-transform method, the histogram of the accepted AR samples closely follows the true parabolic density, confirming a correct implementation.
- ii) From the R run (see console output): empirical acceptance rate = 0.6704 based on 7458 proposals used to obtain 5000 accepted samples, versus the theoretical rate $1/M \approx 0.6667$ (expected ≈ 7500 proposals). The values agree closely, with the small difference explained by simulation randomness.
- iii) Comparing d) and f): the inverse-transform method needs exactly n draws (one uniroot call per sample) and never wastes a draw, but each draw requires solving a root-finding problem numerically, which has computational overhead and needs a valid, invertible CDF. The AR method is simpler to implement (only requires evaluating f and bounding it by Mg) and is easy to vectorise, but it wastes roughly $1 - 1/M \approx 33\%$ of the proposals here. In general, inverse transform is preferable when F^{-1} (even numerically) is cheap to evaluate and no efficient bound M is obvious, while AR is preferable when f is easy to evaluate and bound but hard or impossible to invert, or when a convenient proposal with a small M (close to 1) is available.

Problem 2

a) MLE of β

For $X_1, \dots, X_n \sim \text{Exp}(\beta)$, i.i.d., with pdf

$f(x; \beta) = \frac{1}{\beta} e^{-x/\beta}$, the likelihood and log-likelihood are:

$$L(\beta) = \prod_{i=1}^n \frac{1}{\beta} e^{-x_i/\beta} = \beta^{-n} \exp\left(-\frac{1}{\beta} \sum_{i=1}^n x_i\right)$$

$$\ell(\beta) = \ln L(\beta) = -n \ln \beta - \frac{1}{\beta} \sum_{i=1}^n x_i$$

Differentiating with respect to β and setting equal to zero:

$$\frac{d\ell}{d\beta} = -\frac{n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n x_i = 0$$

which gives $\sum_{i=1}^n x_i = n\beta$, so the MLE is:

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

The second derivative $\ell''(\beta) = n/\beta^2 - 2\sum x_i/\beta^3$ evaluated at $\hat{\beta} = \bar{X}$ is negative, confirming this is a maximum.

b) Bias and variance of the MLE

i) Unbiasedness: for $\text{Exp}(\beta)$, $E(X_i) = \beta$, so:

$$E(\hat{\beta}) = E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \cdot n\beta = \beta$$

Hence $\hat{\beta}$ is an unbiased estimator of β .

ii) Variance: for $\text{Exp}(\beta)$, $\text{Var}(X_i) = \beta^2$. Since the X_i are independent:

$$\text{Var}(\hat{\beta}) = \text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\beta^2 = \frac{\beta^2}{n}$$

c) Sample size for a margin of error of 1 mm (95% confidence)

By the CLT, for large n , $\hat{\beta} = \bar{X}$ is approximately normally distributed, $\hat{\beta} \approx N(\beta, \beta^2/n)$. We want the margin of error of a 95% interval to be at most 1 mm:

$$1.96 \cdot \text{SE}(\hat{\beta}) = 1.96 \cdot \frac{\beta}{\sqrt{n}} \leq 1$$

which gives:

$$n \geq (1.96 \beta)^2 = 3.8416 \beta^2$$

Since β is unknown, we plug in the sample estimate $\hat{\beta} = \bar{x} \approx 7.845$ mm from the data in Problem 2 (computed in part d):

$$n \geq 3.8416 \cdot (7.845)^2 \approx 3.8416 \cdot 61.54 \approx 236.4 \Rightarrow n = 237$$

So we would need approximately 237 wet-day measurements (rounding up) to be 95% confident that $\hat{\beta}$ is within 1 mm of the true β , using the current estimate of β as a planning value.

d) [R] MLE and CI from the rain data - see mandatory1.R

i) The MLE computed in R is $\hat{\beta} = \bar{x} = 7.845$ mm ($n = 40$ observations).

ii) Approximate 95% CI using the CLT, $\hat{\beta} \pm 1.96 \cdot \hat{\beta}/\sqrt{n}$ (plugging in $\hat{\beta}$ for the unknown β in the standard error): the R output gives the interval (5.414, 10.276) mm. Interpretation: we are approximately 95% confident that the true mean wet-day rainfall β lies between about 5.4 mm and 10.3 mm; equivalently, if we repeated this sampling procedure many times, about 95% of such intervals would contain the true value of β .

Problem 3

a) [R] ECDF of rain vs. fitted $\text{Exp}(\hat{\beta})$ CDF

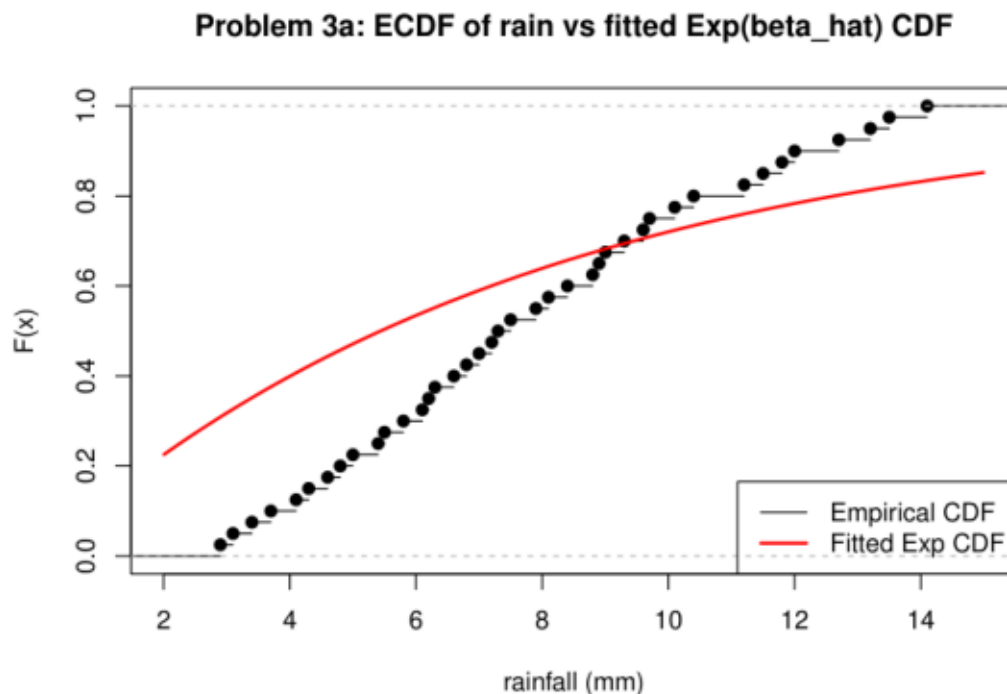


Figure 4: ECDF of the rain data vs. the CDF of the fitted $\text{Exponential}(\hat{\beta})$ model.

b) [R] Histogram with fitted Exponential density and KDE

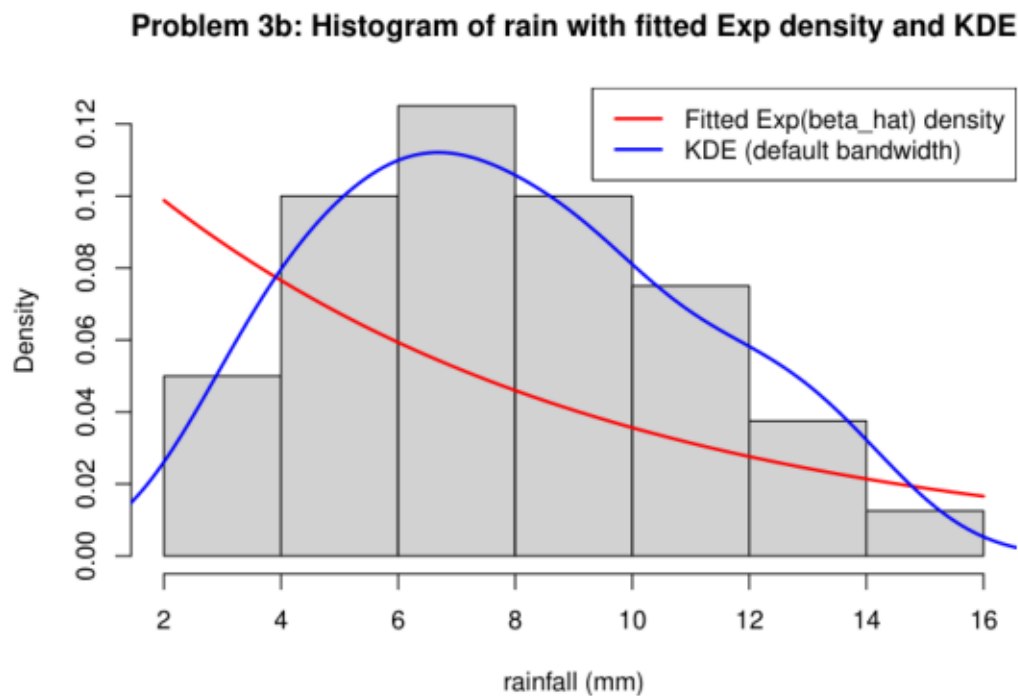


Figure 5: histogram of rain (Scott's rule) with fitted $\text{Exp}(\hat{\beta})$ density and default-bandwidth KDE overlaid.

c) Assessing the Exponential fit and an alternative model

i) Does the Exponential distribution fit well? No. Figure 4 shows a clear systematic discrepancy between the empirical CDF and the fitted exponential CDF: the empirical CDF is S-shaped (concave near the smallest values, then convex, then concave again), while the exponential CDF is concave everywhere and rises very fast from $x = 0$. In Figure 5 the fitted exponential density is monotonically decreasing and largest near $x = 0$, whereas the histogram/KDE clearly show a hump with a mode around 6–8 mm and almost no mass near zero. This mismatch happens because the exponential distribution always has its mode at 0, but the rainfall data is unimodal with a mode clearly away from 0, so an exponential model is a poor fit.

ii) A better alternative: a Gamma distribution. The Gamma distribution generalises the Exponential (which is the special case with shape parameter 1) by adding a shape parameter that allows a mode away from zero and a flexible, unimodal right-skewed shape – exactly the pattern seen in the histogram of the rain data.

d) [R] Fitting a Gamma model and comparing bandwidths – see mandatory1.

Fitting via `fitdistr(rain, "gamma")` gives $\hat{\text{shape}} \approx 6.225$ and $\hat{\text{rate}} \approx 0.7935$ (implied mean = $\text{shape}/\text{rate} \approx 7.845$ mm, matching the sample mean, as expected).

Problem 3d: Fitted Gamma density and KDE at three bandwidths

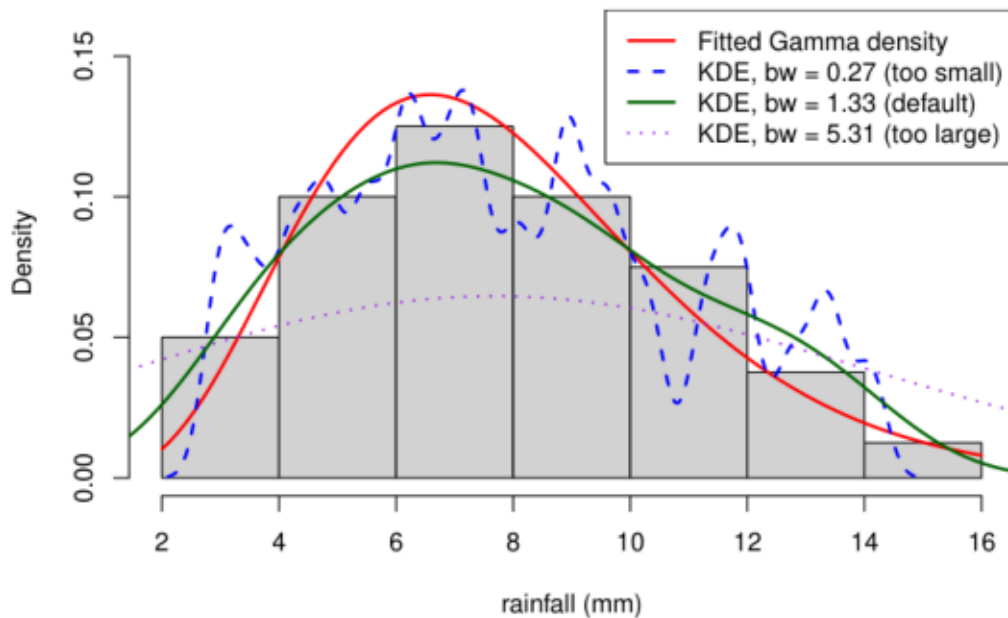


Figure 6: histogram of rain with the fitted Gamma density and KDEs using an undersmoothed, default ($bw.nrd0$), and oversmoothed bandwidth.

Comparison and bias-variance tradeoff: the fitted Gamma density is smooth and matches the overall shape of the histogram and the default-bandwidth KDE closely – both capture a single mode around 6–8 mm and the right-skewed tail. The too-small bandwidth KDE is very wiggly, showing many spurious bumps that are artefacts of individual observations rather than real features of the underlying distribution: this is a low-bias, high-variance estimate. The too-large bandwidth KDE is almost flat and considerably oversmooths the data, losing the mode entirely and pulling mass out into regions with little data: this is a high-bias, low-variance estimate. The default bandwidth ($bw.nrd0$) sits between these extremes and, like the parametric Gamma fit, gives a reasonable compromise – illustrating the classical bias-variance tradeoff in bandwidth selection: small bandwidths chase noise in the sample (low bias, high variance), while large bandwidths smooth out genuine structure (high bias, low variance).

Problem 4

a) Expected value and standard deviation of $Y = A+B+C+D$

Since A, B, C, D are independent, $E(Y)$ is the sum of the means and $\text{Var}(Y)$ is the sum of the variances. Using the given distributions:

$$E(A) = 400, \quad \text{Var}(A) = 900$$

$$E(B) = 250, \quad \text{Var}(B) = \beta_B^2 = 250^2 = 62500$$

$$E(C) = \frac{50+150}{2} = 100, \quad \text{Var}(C) = \frac{(150-50)^2}{12} = \frac{10000}{12} \approx 833.33$$

$$E(D) = 0.3 \cdot 80 = 24, \quad \text{Var}(D) = 80^2 \cdot 0.3 \cdot 0.7 = 6400 \cdot 0.21 = 1344$$

Summing (B is Exponential with mean $\beta_B = 250$, so $\text{Var}(B) = \beta_B^2$):

$$E(Y) = 400 + 250 + 100 + 24 = 774 \text{ mm}$$

$$\text{Var}(Y) = 900 + 62500 + 833.33 + 1344 = 65577.33$$

$$\text{SD}(Y) = \sqrt{65577.33} \approx 256.08 \text{ mm}$$

b) [R] Simulation of Y (n=5000) - see mandatory1.R

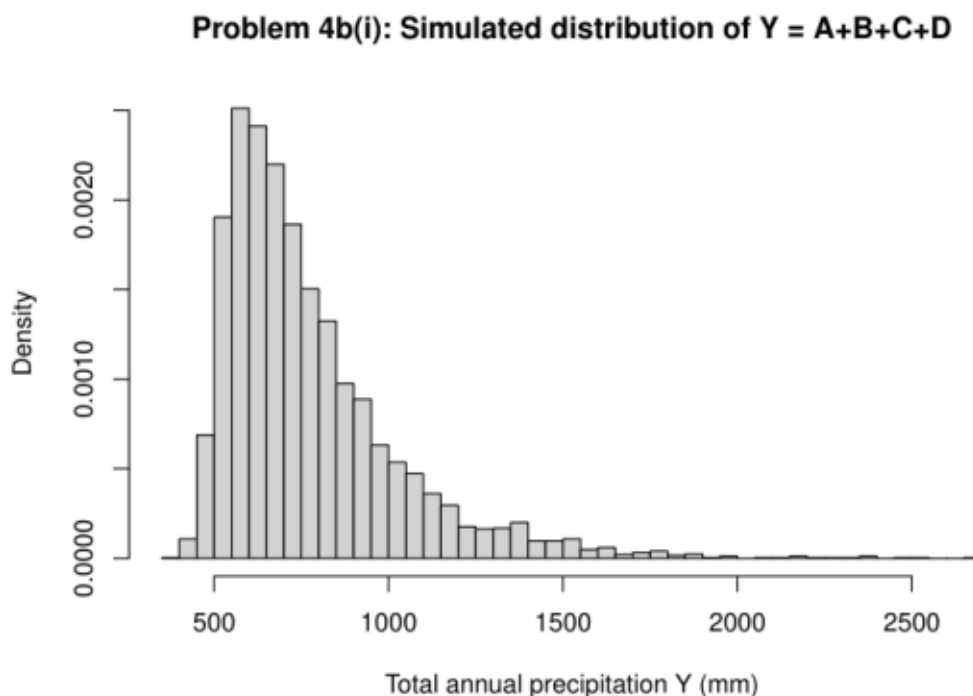


Figure 7: histogram of $n=5000$ simulated values of $Y = A+B+C+D$.

- i) See Figure 7: the simulated distribution of Y is unimodal and right-skewed (the skewness is inherited mainly from the Exponential component B, which has by far the largest variance among the four sources).
- ii) From the R run: simulated $\text{mean}(Y) = 773.01$ and simulated $\text{sd}(Y) = 255.44$, versus the theoretical

values $E(Y) = 774$ and $SD(Y) \approx 256.08$ from part a). The simulated and analytical values agree closely, with the small remaining differences due to Monte Carlo sampling variability for $n = 5000$ replications.

iii) The simulated estimate of $P(Y > 1050)$ is 0.1224 (i.e. about 12.2%). Interpretation: there is roughly a 1-in-8 chance that total annual precipitation at the Stavanger station exceeds 1050 mm in a given year, according to this model; this probability could be estimated because Y 's exact analytical distribution (a sum of Normal, Exponential, Uniform and a two-point/Bernoulli-scaled variable) has no simple closed form, so Monte Carlo simulation is a convenient way to approximate such tail probabilities.

Sources

STA510 Statistical modelling and simulation, autumn 2026 – lecture notes and course R-examples (in particular the inverse-transform and acceptance-rejection example scripts, and the kernel density estimation examples), used as the basis for the simulation methods and R code style applied above. R documentation for `uniroot()`, `density()`, `bw.nrd0()`, and `fitdistr()` (MASS package).